Predictive Maintenance for Aerospace Systems

Cheng (Tony) Ke, Felipe Villanueva, Yicun (Ethan) Wu

Department of Electrical and Computer Engineering
University of Rochester
Rochester, New York, USA

ABSTRACT

Predictive maintenance has become increasingly important in aerospace systems due to the high cost and safety risks associated with aircraft engine failures. This project investigates Remaining Useful Life (RUL) prediction using the NASA CMAPSS FD001 turbofan engine dataset, which contains multivariate time-series sensor measurements collected throughout engine operation. Time-domain and frequency-domain learning approaches were explored by comparing Linear Regression, 1D Convolutional Neural Networks (1D CNN), 2D Convolutional Neural Networks (2D CNN) using Short-Time Fourier Transform (STFT) spectrograms, and Long Short-Term Memory (LSTM) networks. Bidirectional Long Short-Term Memory (BiLSTM) was also evaluated as an exploratory extension. The dataset was processed using sliding windows of 30 timesteps to capture degradation behavior over time. Experimental results showed that the LSTM achieved the strongest performance by effectively learning long-term temporal dependencies within engine degradation patterns, outperforming both convolutional and baseline approaches. Overall, this work demonstrates the effectiveness of deep learning for predictive maintenance applications and highlights the importance of temporal modeling for aerospace engine health monitoring.

Predictive Maintenance, RUL Prediction, LSTM, CNN, Aerospace Systems, Deep Learning

1. INTRODUCTION

Aircraft engines generate large amounts of sensor data during operation, providing valuable information about engine health and degradation behavior. Unexpected engine failures can lead to significant safety risks, operational downtime, and increased maintenance costs. Traditional maintenance strategies are often reactive or based on fixed service schedules, which may result in unnecessary inspections or failures occurring before maintenance can be performed. As a result, predictive maintenance has become an important research area in aerospace engineering because it enables earlier detection of degradation and more efficient maintenance planning. One of the key goals of predictive maintenance is estimating the Remaining Useful Life (RUL) of an engine, which refers to the amount of operational time remaining before failure occurs. Accurate RUL prediction allows maintenance to be scheduled proactively, improving reliability while reducing costs. However, predicting RUL is challenging because engine degradation is nonlinear, time-dependent, and influenced by multiple interacting sensor measurements. In this project, the NASA CMAPSS FD001 turbofan engine dataset was used to investigate predictive maintenance for aerospace engines. The dataset contains multivariate time-series sensor measurements collected from engines operating until failure. Sensor data was processed into sliding windows of 30 timesteps to capture temporal degradation behavior. Several machine learning and deep learning approaches were explored, including Linear Regression, 1D Convolutional Neural Networks (1D CNN), 2D Convolutional Neural Networks (2D CNN) using STFT spectrograms, and Long Short-Term Memory (LSTM) networks. The objective of this work is to compare time-domain and frequency-domain learning approaches for RUL prediction and determine which model provides the most accurate and reliable performance.

2. METHODS

The NASA CMAPSS FD001 dataset was selected because it provides multivariate time-series sensor measurements from turbofan engines operating under a single operating condition and fault mode (High Pressure Compressor). Each engine contains 24 sensor measurements recorded over multiple operational cycles until failure occurs. Remaining Useful Life (RUL) values were computed for each cycle by subtracting the current cycle from the engine's final

failure cycle.

2.1. Dataset

The NASA CMAPSS FD001 dataset was selected because it provides multivariate time-series sensor measurements from turbofan engines operating under a single operating condition and fault mode. Each engine contains 24 sensor measurements recorded over multiple operational cycles until failure occurs. Remaining Useful Life (RUL) values were computed for each cycle by subtracting the current cycle from the engine's final failure cycle.

2.2 Data Prerocessing

Sensor measurements were normalized and organized into sliding windows of 30 timesteps. Each sliding window represents a short sequence of engine behavior and serves as one training sample. This approach allows the models to learn temporal degradation patterns over time rather than relying on individual sensor measurements. The dataset was divided into training and validation subsets for evaluation.

2.3 Linear Regression

Linear Regression was implemented as a baseline model to evaluate the limitations of simple statistical methods for RUL prediction. Sensor sequences were flattened into feature vectors before training.

2.4 1D CNN

A 1D Convolutional Neural Network (1D CNN) was implemented to learn local temporal features directly from raw time-series sensor data. Convolutional layers enabled the model to identify degradation-related patterns across sequential sensor measurements.

2.5 2D CNN with STFT

A 2D Convolutional Neural Network (2D CNN) was developed using Short-Time Fourier Transform (STFT) spectrograms. The STFT transformed time-series sensor signals into time-frequency representations, allowing the model to learn spatial degradation patterns from spectral information.

2.6 LSTM

A Long Short-Term Memory (LSTM) network was implemented to capture long-term temporal dependencies and sequential degradation behavior in engine sensor data. Memory cells within the LSTM allowed the model to retain important historical information across multiple operational cycles.

2.7 Linear Regression

A Bidirectional Long Short-Term Memory (BiLSTM) network was explored as an extension of the standard LSTM architecture. The BiLSTM processes sequential information in both forward and backward directions to potentially capture richer temporal context within degradation sequences.

3. EXPERIMENT

All neural network models were trained using the Adam optimizer and Mean Squared Error (MSE) loss function for 25 epochs. Model performance was evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and coefficient of determination (R^2). Predicted versus actual RUL visualizations were also analyzed to evaluate each model's ability to capture degradation trends and generalize across different engines.

*A cycle represents one operational time step in the simulated engine dataset.

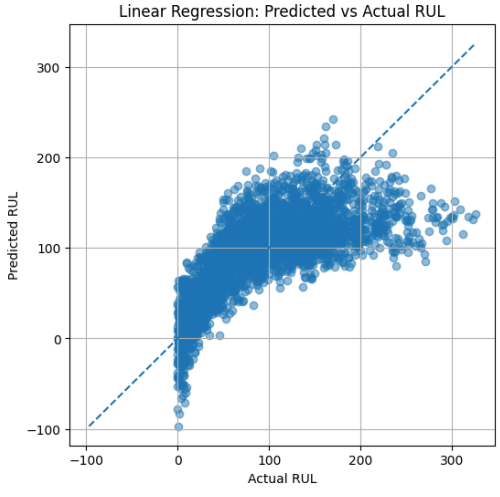


Fig. 1. Linear Regression Plot

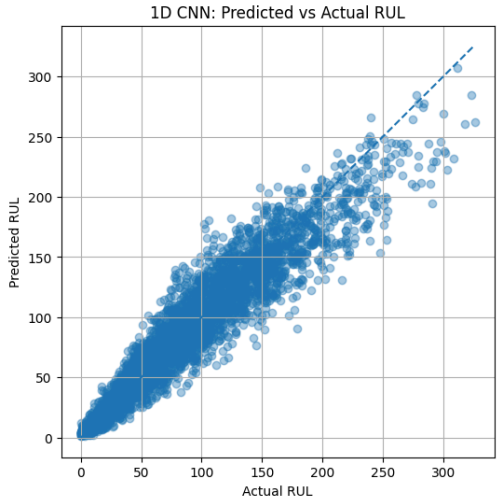


Fig. 2. 1D CNN Plot

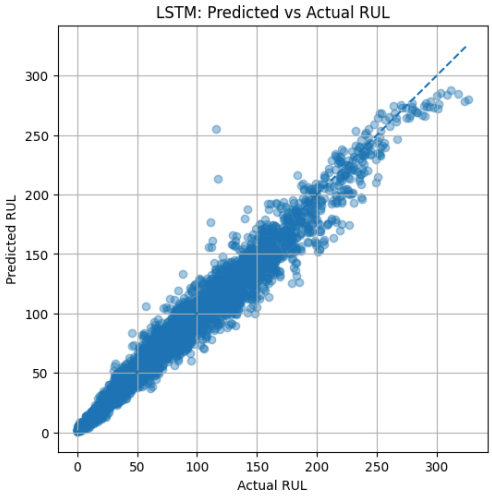


Fig. 4. 2D CNN Plot

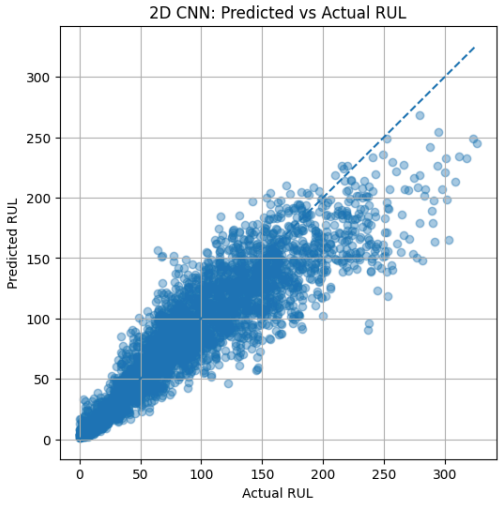


Fig. 3. 2D CNN Plot

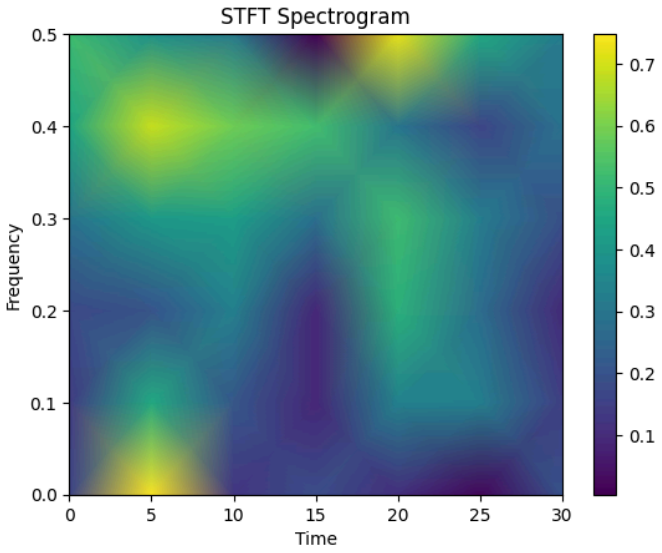


Fig. 4a. 2D CNN Spectrogram

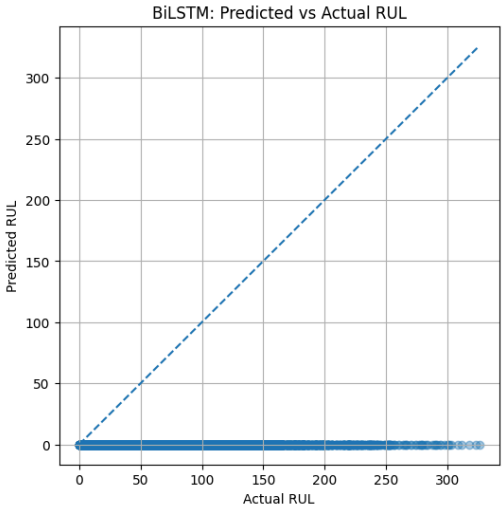


Fig. 5. BiLSTM Plot

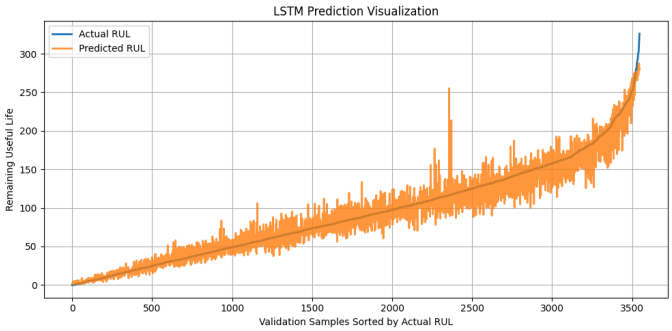


Fig. 7. LSTM Predition

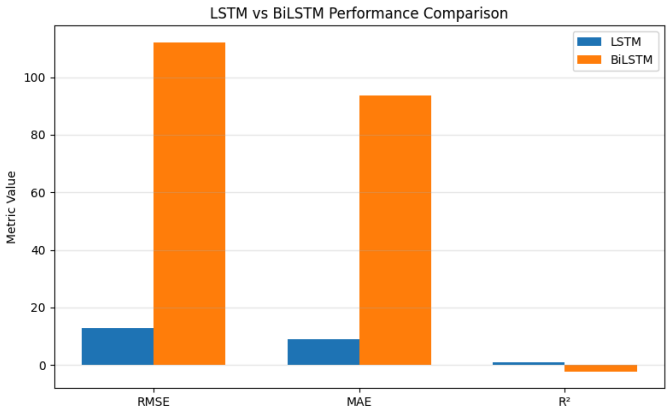


Fig. 6. LSTM vs BiLSTM Bar Graph

4. RESULTS

The LSTM achieved the strongest overall performance among all evaluated models. The model produced the lowest RMSE and MAE values while achieving the highest R^2 score, indicating strong alignment between predicted and actual RUL values. The 1D CNN also demonstrated strong predictive capability by learning local temporal features from sequential sensor data. The 2D CNN using STFT spectrograms captured frequency-based degradation behavior but experienced reduced performance due to partial loss of temporal resolution. Linear Regression produced the weakest performance because it could not effectively model nonlinear temporal degradation behavior.

Model	MSE	RMSE	MAE	R^2
Linear Regression	1408.42	37.53	28.14	0.63
1D CNN	398.36	19.96	14.55	0.90
2D CNN	738.21	27.17	20.29	0.81
LSTM	162.32	12.74	8.66	0.96
BiLSTM	12544.00	112.00	94.00	-2.00

5. DISCUSSION

The experimental results demonstrate that deep learning models are significantly more effective than traditional regression methods for Remaining Useful Life (RUL) prediction in aerospace engines. Linear Regression served as a useful baseline model; however, its inability to model nonlinear temporal degradation behavior resulted in poor predictive performance and visible underfitting in the predicted versus actual RUL plots. The spread and curvature of the predictions indicated that simple statistical relationships are insufficient for capturing the complex degradation dynamics present in turbofan engine sensor data. The 1D CNN showed substantial improvement over Linear Regression by learning local temporal features directly from the raw sensor sequences. Convolutional layers were able to identify short-term degradation patterns and local trends in the data, leading to lower prediction error and stronger alignment with the ideal prediction line. However, while the 1D CNN performed well, it still had limitations in capturing long-range temporal dependencies across the entire engine lifecycle. The 2D CNN experiment provided insight into the advantages and disadvantages of spectral learning approaches. By transforming the sensor data into STFT spectrograms, the model was able to learn time-frequency representations similar to image processing tasks. Although the 2D CNN successfully identified some degradation characteristics, the conversion process reduced temporal resolution and likely removed important sequential information contained in the original time-series data. As a result, performance was lower than the time-domain approaches, particularly the LSTM model. Among all evaluated models, the LSTM achieved the strongest performance. The predicted versus actual RUL visualizations showed the tightest clustering around the ideal prediction line, and the evaluation metrics demonstrated the lowest prediction error and highest coefficient of determination. This result highlights the importance of temporal memory in predictive maintenance applications. Since engine degradation is inherently sequential and evolves gradually over time, the LSTM's ability to retain and update information through memory cells allowed it to capture long-term degradation trends more effectively than convolutional or statistical methods. The BiLSTM extension provided additional insight into model complexity and dataset suitability. Although Bidirectional LSTMs are capable of learning temporal information in both forward and backward directions, the model performed poorly compared to the standard LSTM. One possible explanation is that the FD001 dataset contains relatively structured degradation behavior under a single operating condition, meaning that forward temporal information alone was sufficient for effective prediction. Additionally, the increased complexity of the BiLSTM may have contributed to optimization difficulties or overfitting. Overall, the experiments emphasize that temporal modeling plays a critical role in predictive maintenance for aerospace systems. The comparison between time-domain and frequency-domain approaches demonstrates that preserving sequential information is essential for accurate RUL prediction. These findings suggest that advanced sequence-based architectures such as LSTMs, Transformers, or hybrid temporal models may

provide even stronger predictive performance for future aerospace prognostics research.

6. CONCLUSSION

This project demonstrated that machine learning and deep learning methods can effectively predict the Remaining Useful Life (RUL) of aerospace engines using multivariate sensor data from the NASA CMAPSS FD001 dataset. Among all evaluated models, the LSTM achieved the best overall performance by effectively capturing long-term temporal dependencies within engine degradation sequences. Overall, the results emphasize the importance of temporal modeling for predictive maintenance applications and highlight the potential of deep learning for improving reliability and reducing maintenance costs in aerospace systems.

7. REFERENCES

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